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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 4: Current Scenario of Transportation in India

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Contents

- Transportation in India
- Transport problems
- Transport strategies in India
- Non-Motorized Transport in India
- Pre and Post Covid Public Transport



Transportation in India

- As per the 2011 census, 31.2% of India's population (377 million) is living in urban areas.
- As the UN estimates, these numbers will grow to 40% (590 million) by 2030 and 58% (875 million) by 2050.
- Approximately 63% of India's Gross Domestic Product (GDP) is contributed by these 30% population in urban areas.
- Scope for PURA (Shri APJ Abdul Kalam)



Source: (Intelligent Transport, 2016)

Transport problems in India



Severe congestion



Growth in number of private vehicles





Increasing Greenhouse gas (GHG) emissions



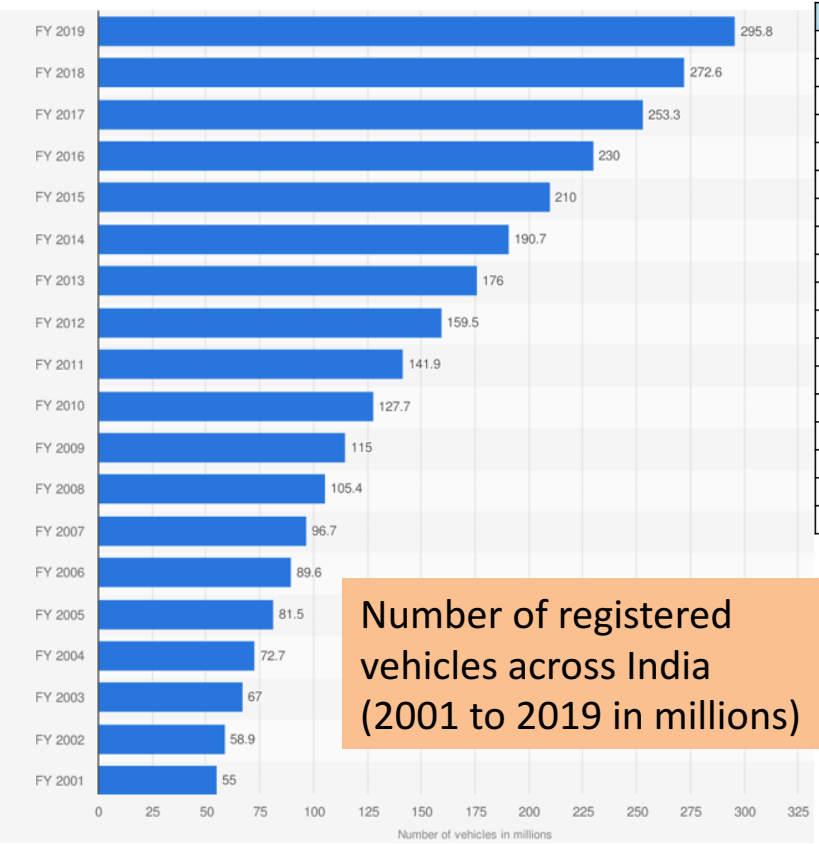
Deteriorating Air quality

Increasing Road accidents



Source: (Intelligent Transport, 2016)

Transport problems in India: Growth of Private-vehicle ownership



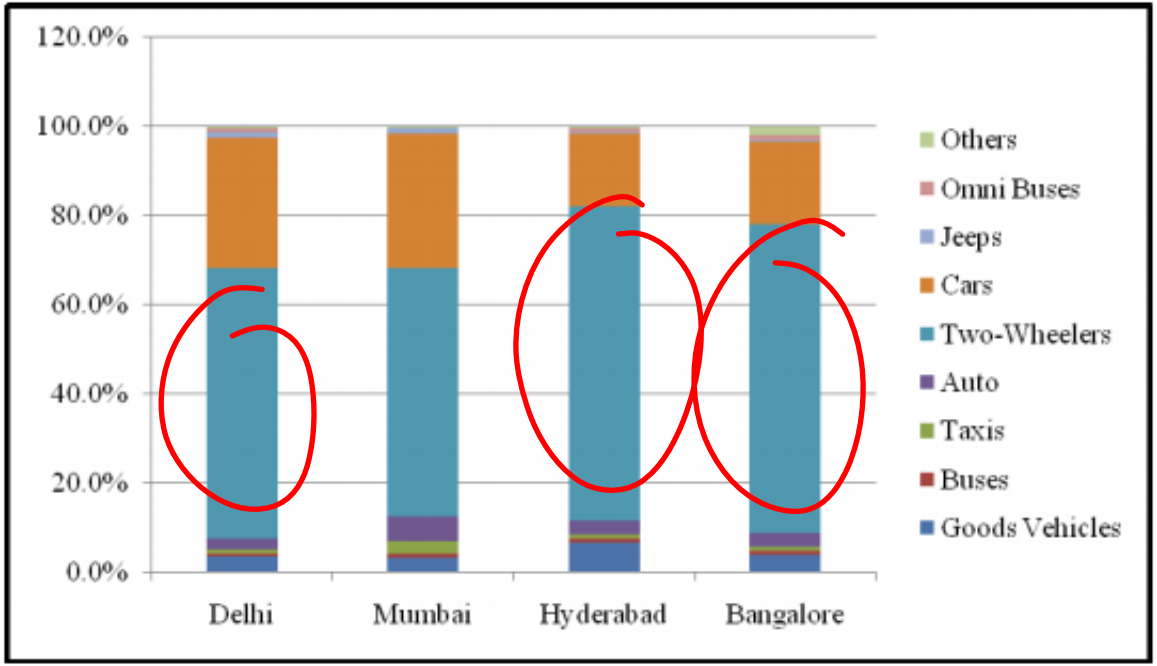
Number of registered vehicles across India (2001 to 2019 in millions)

Year	Yearly Growth Rate (%)
2002	7.1%
2003	13.7%
2004	8.5%
2005	12.1%
2006	9.9%
2007	7.9%
2008	9%
2009	9.1%
2010	11%
2011	11.12%
2012	12.4%
2013	10.34%
2014	8.35%
2015	10.12%
2016	9.52%
2017	10.13
2018	7.62%
2019	8.51%

- An Average annual growth rate of 10%.
- The private vehicles ownership have increased to 438% in 2019, compared to the base year 2001.

Source: (Ministry of Road Transport and Highways (India), 2021)

Share of Types of Motorized Vehicles in India (2011)



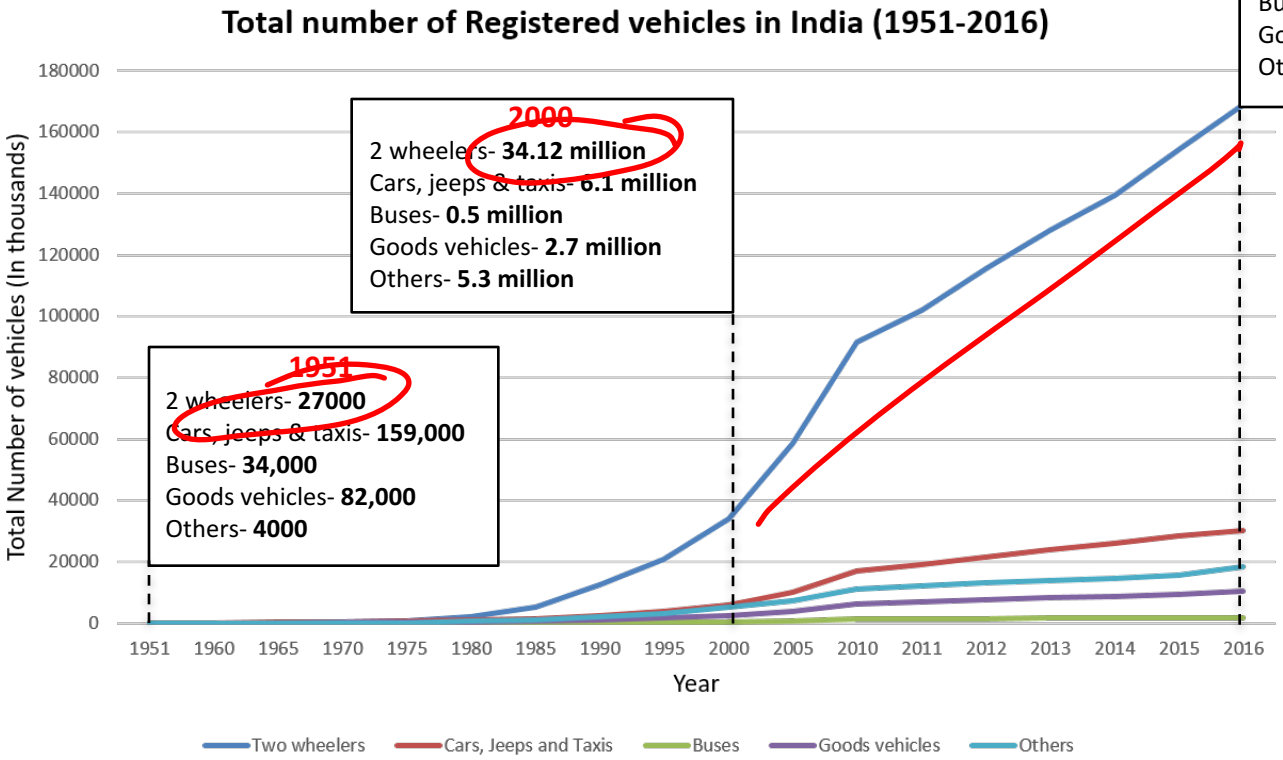
Source: Ministry of Road Transport, Government of India

City-wise ownership of Motorized vehicles in India, 2011

- Two wheelers have the most share (about 75%) in Indian transportation followed by private cars.
- Two-wheeler ownership have doubled in 2020, compared to 2011.

Source: (UNESCAP 2013)

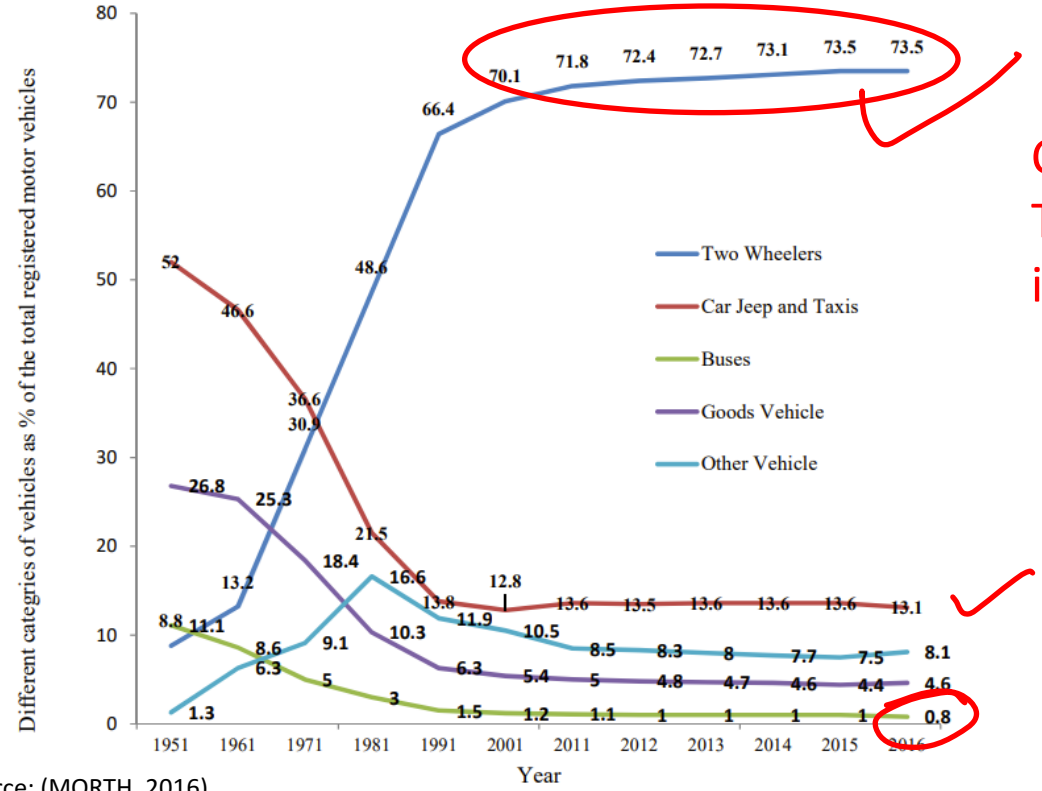
Registered Motor Vehicles in India



- The total number of vehicles increased from 0.3 million in 1951 to 230.03 million in 2016.

Source: (MORTH, 2016)

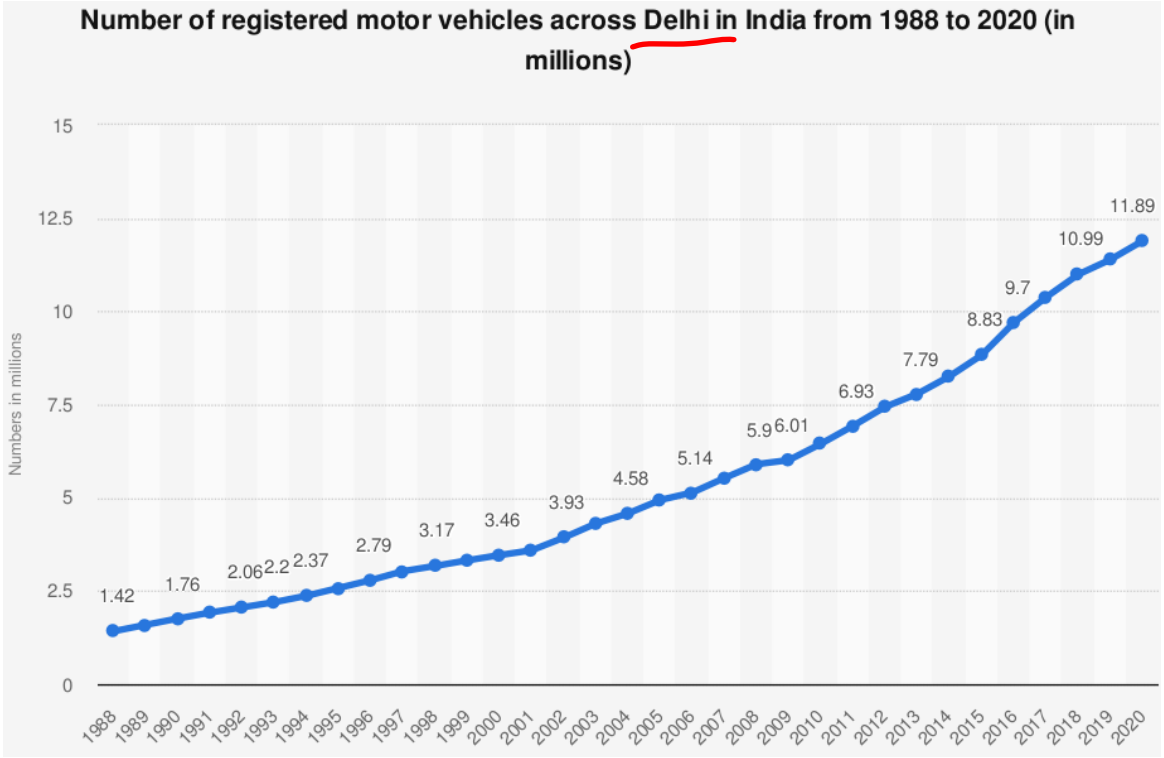
Composition of Registered Motor Vehicles in India (%)



Source: (MORTH, 2016)

Composition (as percentage of Total) of Registered Motor vehicles in India from 1951 to 2016

Transport problems in India: Growth of Private-vehicle ownership



Increase in Vehicle count across Delhi

- Motor vehicle use have doubled during the last decade from 2010 to 2020

Source: (Ministry of Road Transport and Highways (India), 2021)

Transport problems in India: Growth of Private-vehicle ownership

- Poor Transport Infrastructure compared to the vehicle increased
- Registered vehicle increase in the last decade: 219%.
- Urban road infrastructure increase in the last decade: 124%.
- Rapid motorisation has led to severe congestion, longer journeys and higher per capita trips

Source: (Ministry of Road Transport and Highways (India), 2021)

Transport problems in India

Congestion



Traffic congestion in Mumbai Source: (https://trak.in/tags/business/2019/06/05/mumbai-has-worlds-worst-traffic-flow-rs-41000-crore-is-lost-per-year-due-to-insane-traffic/)



Traffic congestion in Delhi Source: (https://www.newsbytesapp.com/news/india/projects-to-ease-traffic-in-delhi/story)



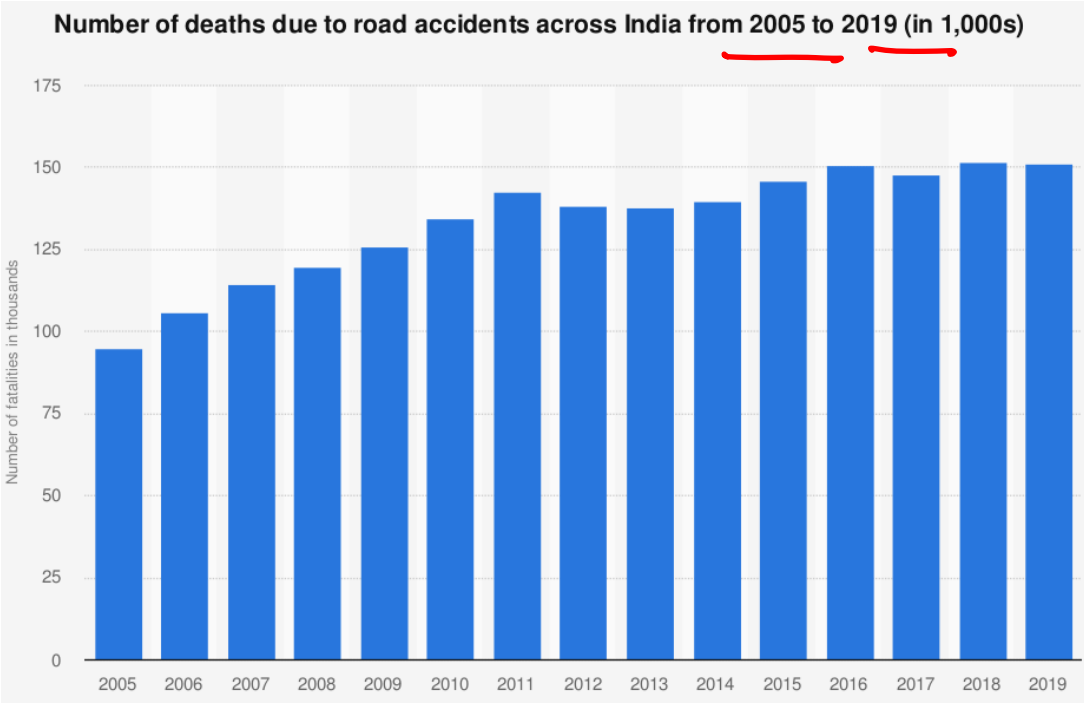
Traffic congestion in Kolkata Source: (https://timesofindia.indiatimes.com/city/kolkata/traffic-chaos-at-salt-lake-vip-road/articleshow/64904089.cms)

- 3 Indian cities in top 10 World Congested cities
- According to Global traffic congestion levels, Mumbai city is ranked 2nd in the world, Bengaluru 6th, Delhi 8th and Pune 16th among 416 cities listed across 56 countries

Source: (https://timesofindia.indiatimes.com/india/mumbai-bengaluru-delhi-in-top-10-world-cities-in-traffic-congestion-indicating-revival-of-economic-activities/articleshow/80246426.cms)

Transport problems in India

Road accidents



- Over 151,000 fatalities due to road accidents annually i.e. approx 414 each day.
- India is accounted for about 6% of the global road traffic incidents.
- Almost 70% of the accidents involves young Indians.

Source: (Ministry of Road Transport and Highways (India), 2021)

Transport problems in India

Road accidents due to Potholes



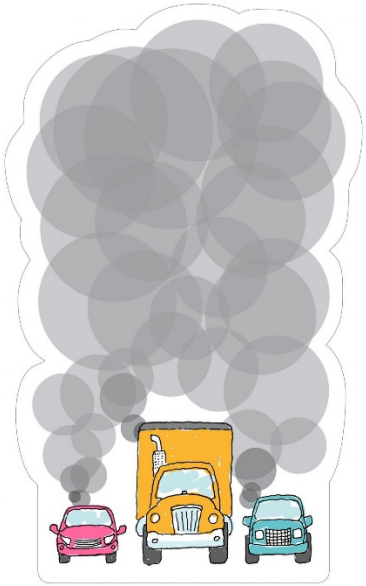
Source: (Ministry of Road Transport and Highways (India), 2021)

- Over 4770 accidents due to potholes in 2019
- One of the major factor causing accidents in India.
- Though the number of road accidents decreased, the number of deaths due to potholes increased significantly in 2019.

Transport problems in India

Air Pollution

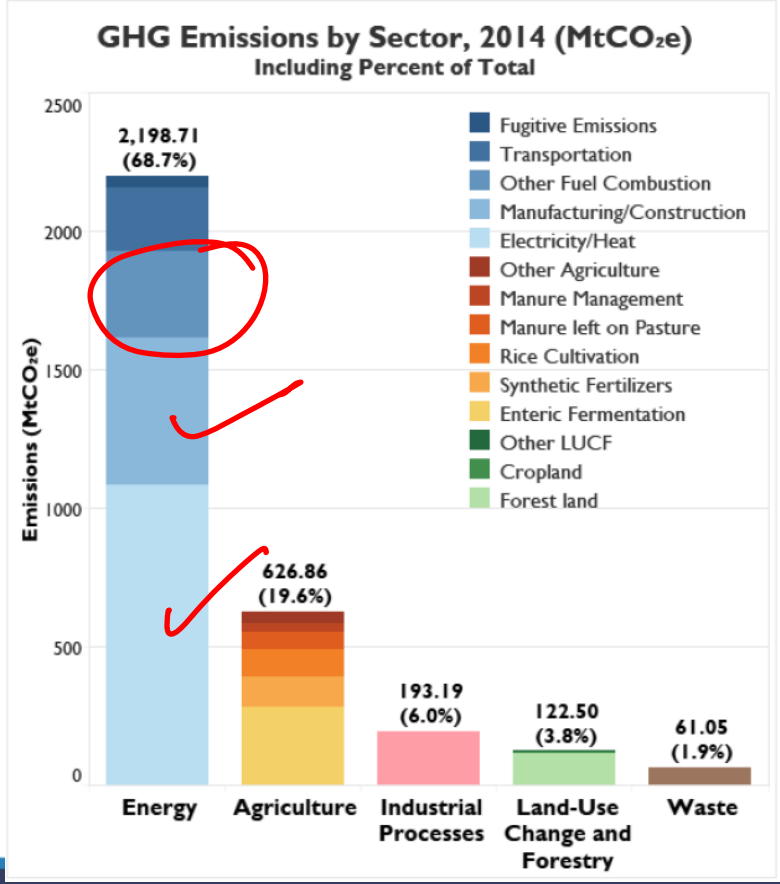
- 4.2 million deaths every year worldwide due to ambient (outdoor) air pollution.
- Transportation is a significant factor in India's air quality problems
- According to the 2017 Global Burden of Disease, around 1.1 million people in India die prematurely each year from diseases directly related to air pollution, making it the fifth leading cause of death in the country.



Source: (WHO, 2018)

Transport problems in India

Air Pollution

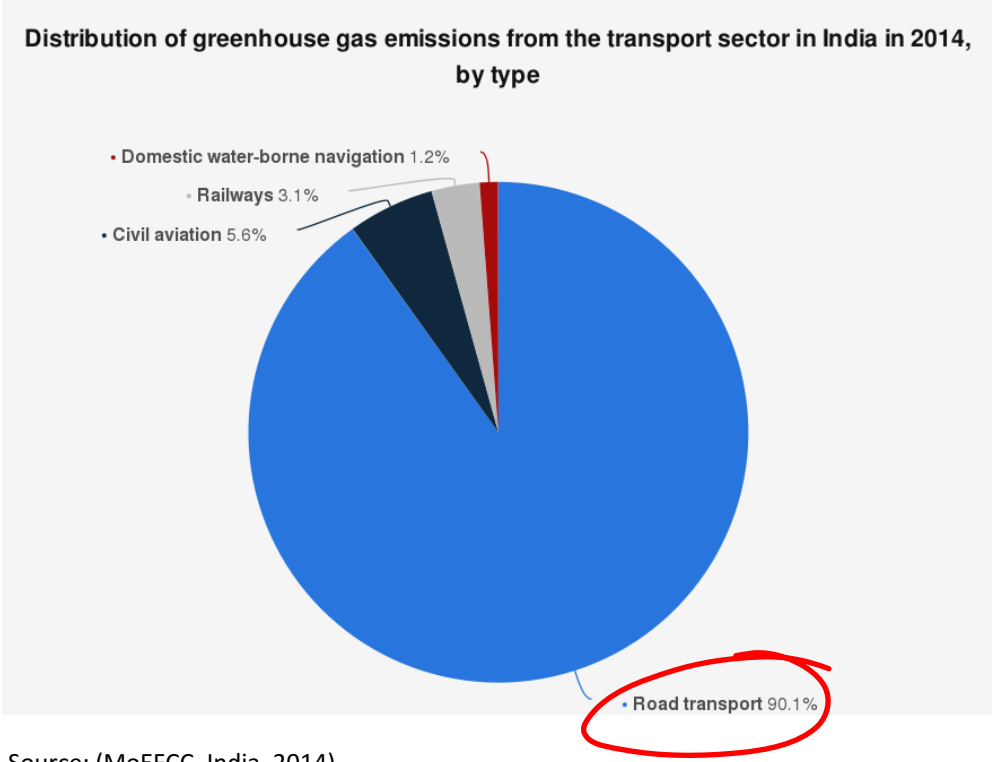


- According to World Resources Institute Climate Analysis Indicators Tool (WRI CAIT) 2018, In India, Energy sector contributes to around 68.7% of the GHG emissions followed by Agriculture (19.6%), Industries (6%), Landuse change and forestry (3.8%) and Waste (1.9%).

Source: (WRI CAIT, 2018)

Transport problems in India

Air Pollution



Source: (MoEFCC, India, 2014)

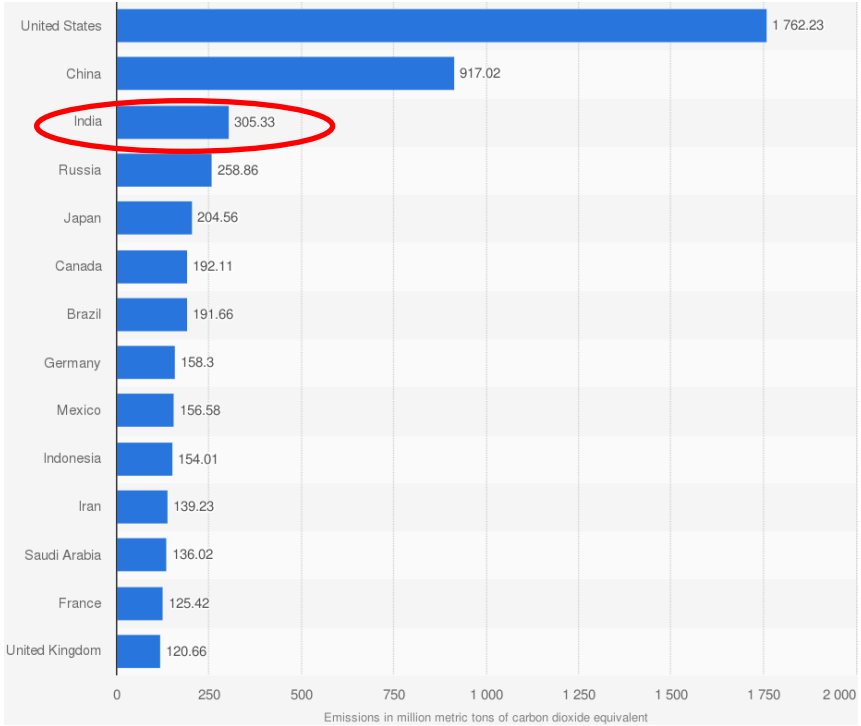
- Road transportation contributes to over 90% of GHG emissions from the Transportation sector in India (2014).

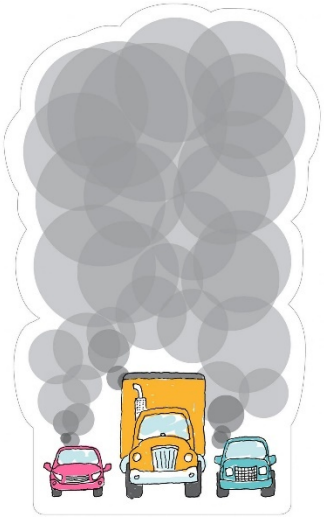
Transport Sector	GHG emissions (Gg CO_2e)
Road Transport	225,155.51
Civil Aviation	14009.68
Railways	7775.36
Water borne Navigation	3002.1
Total	**250172.79 Gg CO_2e**

GHG emission from Transportation sector (2018)

Air Pollution

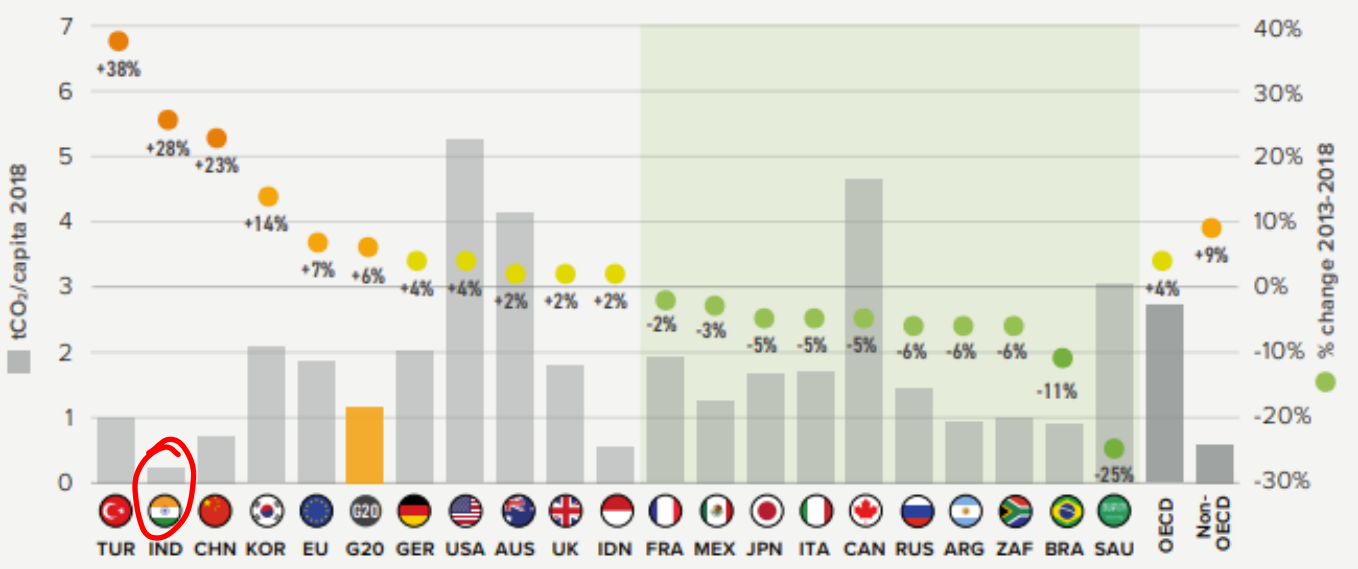
- GHG emissions from the Transportation sector worldwide in 2018 (in million metric tons of CO2 equivalent





Source: (https://www.statista.com/statistics/1084166/ghg-emissions-transportation-sector-globally-by-country/)

Transport emission per-capita in the G20

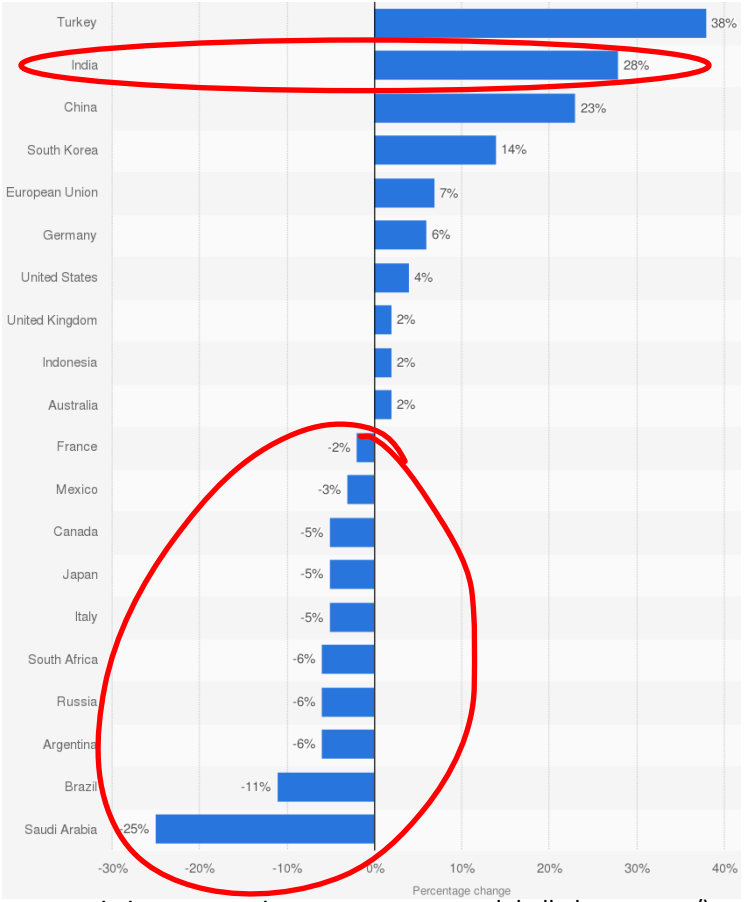


Transportation emission per capita (excluding Aviation) by the G20 countries (2018)

Source: (Climate Transparency Report, 2020)

Transport emission reduction/increase per-capita Worldwide

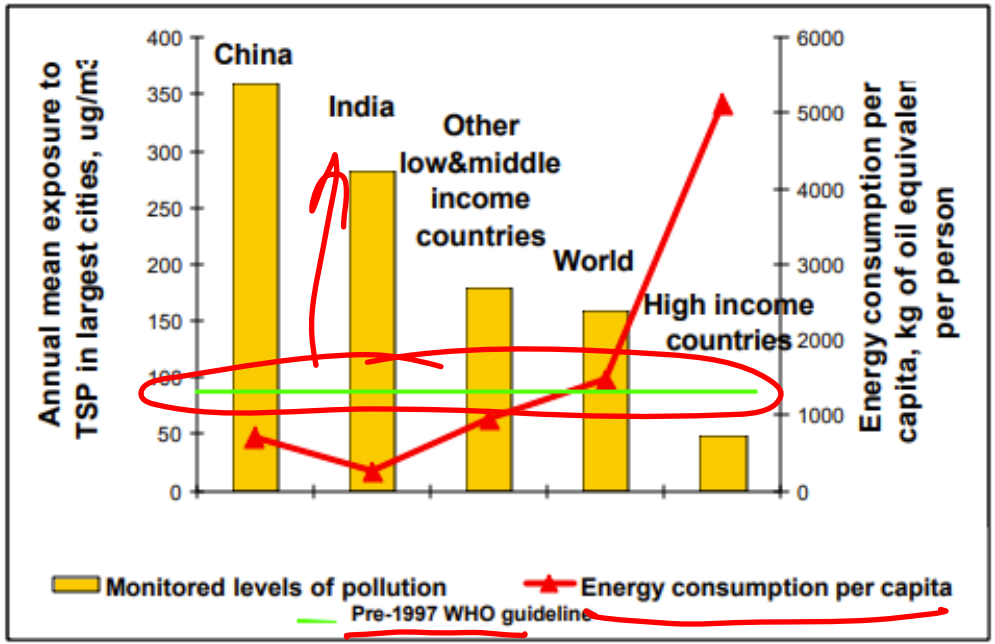
- Emissions in Turkey increased by 38% between 2013 to 2018 among the G20 nations.
- India and China were the 2nd and 3rd largest contributors
- Saudi Arabia reduced transport emissions by 25% from 2013 to 2018.



Source: (https://www.statista.com/statistics/1116633/change-emissions-per-capita-transport-sector-globally-by-country/)

Transport problems in India

Air Pollution



Level of Population's exposure to pollution by world region and per capita energy consumption by world region

Source: (WHO, 2018)

- Transportation sources account for approximately a third of PM pollution in India

Transport problems in India

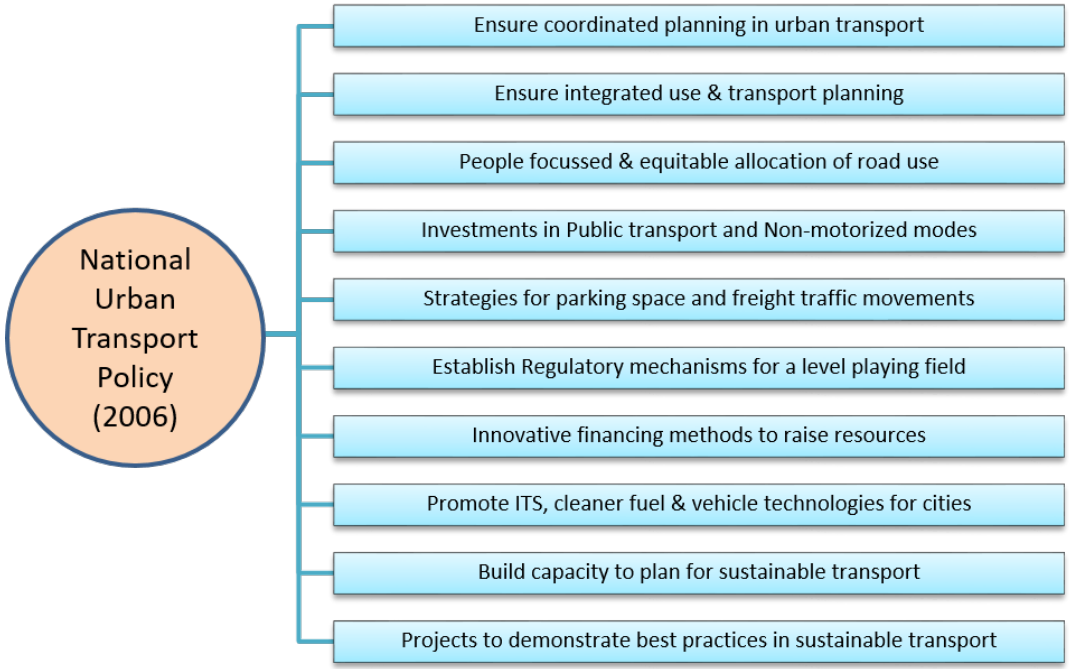
Declining share of Public Transport



- The share of public transport is just 18.1% of the total work trips.
- In 2015 the number of daily trips using a motorcycle for commuting was 35 million (excluding personal trips)
- More people use motorcycles than travel by bus (22.9 million)

Source: (Intelligent Transport, 2016)

National Urban Transport Policy, 2006



- In 2006- The Government of India (GOI) introduced the National Urban Transport Policy (NUTP) for setting the policy framework for providing sustainable mobility for the future.

Source: (Intelligent Transport, 2016)

Transport strategies in India

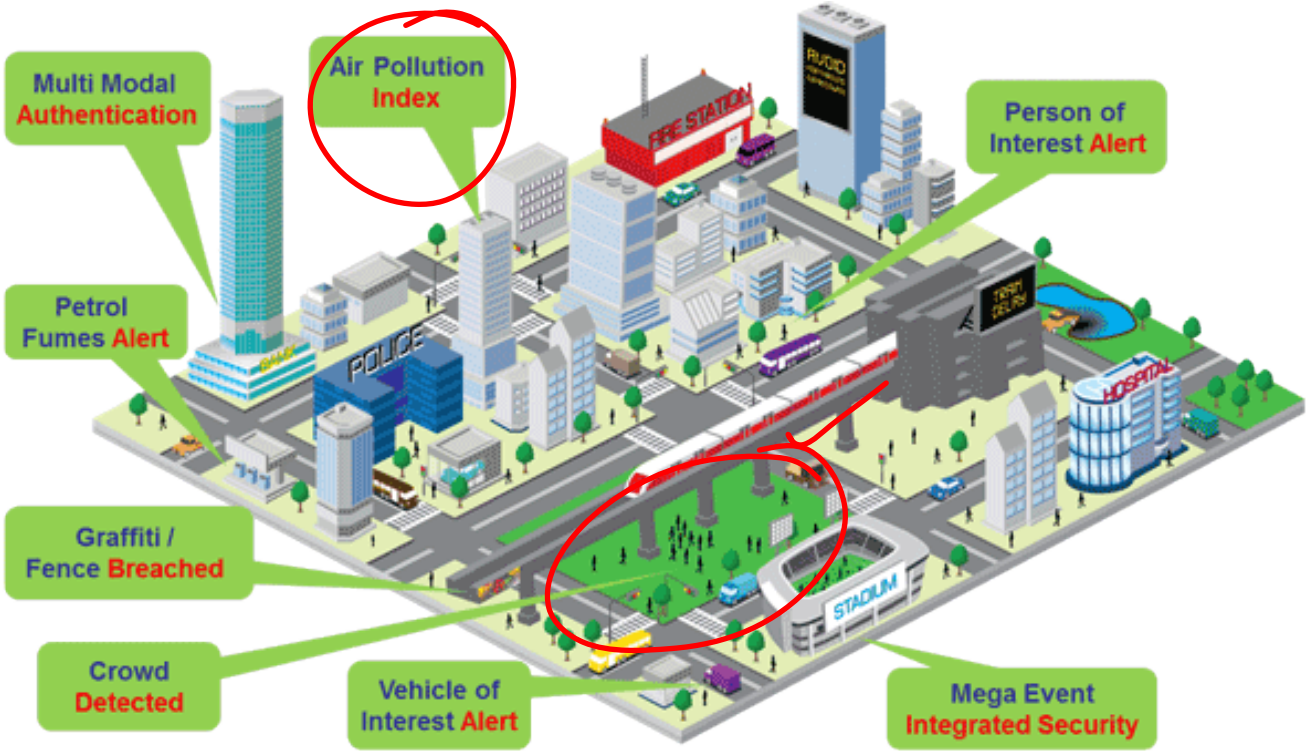
Development of Smart cities



- The Government of India has launched two flagship programmes in 2015.
- 100 Smart Cities and Atal Mission for Rejuvenation, and Urban Transformation (AMRUT) for 500 cities that have a population of 100,000 or more with total funding of $8 billion and $8.3 billion, respectively.
- The smart cities initiatives focus on core infrastructure service, whereas, AMRUT will adopt a project approach to ensure basic infrastructure services.

Source: (Intelligent Transport, 2016)

Transport strategies in India: Development of Smart cities



Source: (Anuj Tiwari and Kamal Jain, esri India)

Transport strategies in India

The emergence of Metro system



- More Indian cities exploring the option of metro system after success of Delhi metro.
- Delhi Metro is the largest metro system with a total length of 213km, carrying 2.8 million passengers a day.
- The Ministry of Urban Development (MoUD) estimates that there is approximately 316km of metro lines currently in operation and more than 500km of metro lines under construction across the country



Source: (Intelligent Transport, 2016)

Transport strategies in India

Revamping City Bus system



- Most popular and convenient mode of transportation in urban cities.
- More than 1.6 million buses are registered in India, and the public bus sector operates 170,000 buses carrying roughly 70 million people per day.
- However, bus transportation has not been able to cater to the growing travel demand.
- Need to procure new vehicles and technology.

Source: (Intelligent Transport, 2016)

Transport strategies in India

Revamping City Bus system

- The GOI launched National Urban Renewal Mission (NURM) in 2009 a massive programme of revitalising urban areas to speed up the creation of much-needed infrastructure.
- Approximately 222km of Bus Rapid Transit (BRT) systems are operational by 2015 and the remaining 282km are under construction.
- MoUD has provided financial assistance to 11 cities for the construction of 504km Bus Rapid Transit System (BRTS).



Source: (Intelligent Transport, 2016)

Transport strategies in India

Future Bus Technology



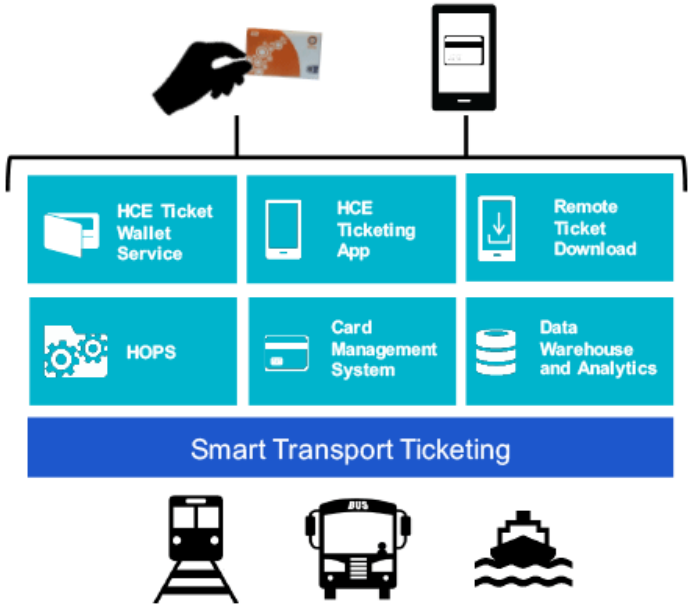
- GOI launched the full indigenous retro-fitted electric buses with hybrid technology, converting existing conventional fuel buses into electric buses.
- Main hindrance to this adoption of new technology was the financial implication and capital cost.
- The average cost of a hybrid bus in India is around INR 23 million, compared to INR 9 million for a premium diesel bus.

Source: (Intelligent Transport, 2016)

Transport strategies in India

Future Bus Technology

- The availability of real-time information is very important for commuters to plan their journey.
- Smart ticketing systems are not yet integrated with the bus system in majority of cities.
- Bengaluru Metropolitan Transport Corporation (BMTC) became the first city in India to launch an open loop EMV contactless smartcard (not integrated with other modes).
- The Mumbai Metropolitan Region Development Authority (MMRDA) has appointed London's Transport for London (TfL) to prepare a detailed plan for an integrated ticketing system in the Mumbai Metropolitan Region.



Source: (Intelligent Transport, 2016)

Non-Motorized Transport in India

- Non-Motorized Transport (NMT) in India mainly includes walking, cycling and cycle rickshaws.

- Green modes of transport
  - Low Carbon footprint
  - Minimal energy consumption
  - Zero local emissions
  - Inexpensive compared to motorized transport







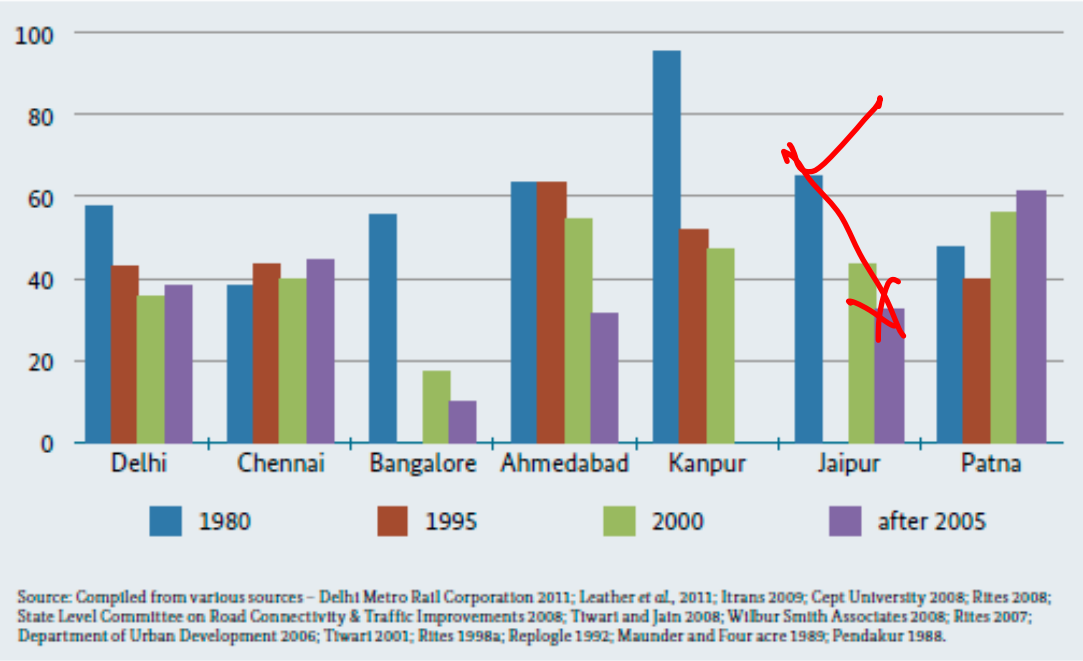
- Low-income households largely depend on NMTs in India

- Has immense health benefits

- NMTs are still neglected as a mobility option in favour of capital and infrastructure intensive modes of transport.

Source: (NTDCP Final Report, 2013)

Non-Motorized Transport in India



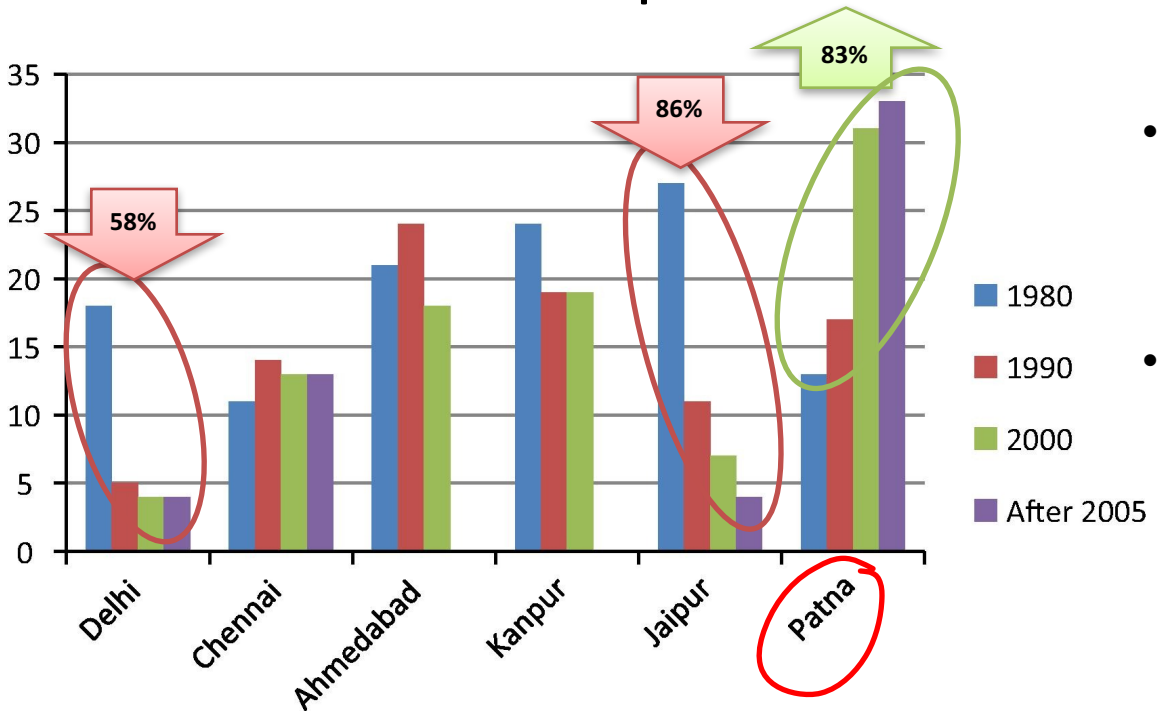
Source: Compiled from various sources – Delhi Metro Rail Corporation 2011; Leather et al., 2011; Itrans 2009; Cept University 2008; Rites 2008; State Level Committee on Road Connectivity & Traffic Improvements 2008; Tiwari and Jain 2008; Wilbur Smith Associates 2008; Rites 2007; Department of Urban Development 2006; Tiwari 2001; Rites 1998a; Replogle 1992; Maunder and Four acre 1989; Pendakur 1988.

Trends in Modal share of NMTs (Walking and Bicycle) since the 1980s

- Share of NMTs (walking and cycling combined) was around 40-60% in Indian cities in 1980s.
- Share of NMTs was declining in Indian cities since the 1980s, except for certain cities such as Chennai and Patna (from graph shown).

Source: (SUTP, 2015)

Non-Motorized Transport in India



Trends in Modal share of Bicycle trips, since the 1980s

- Bicycle trips accounted for approx. 10-30% of the total trips.
- Bicycle trips declined by 22% in Kanpur by 1995, compared to 58% in Delhi and 86% in Jaipur.
- Bicycle share increased in Patna by 83% between 1990s and 2000s and again by 6% by 2008.

Source: (SUTP, 2015)

Non-Motorized Transport in India

- NMTs blamed for congestion issues in roads
- NMTs such as Rickshaws considerably increases road capacity on segregated lanes
- However, 24% of Delhi's metro trips are dependant on Rickshaws as feeder mode.





Source: (SUTP, 2015)

Non-Motorized Transport in India

Key Policy Initiatives by MoUD in India



- Vision of the National Urban Transport Policy (NUTP) states that NMT should be the last mile connector for urban transport system and for short independent trips.

- The first two principles of the National Mission for Sustainable Habitat, under the Prime Minister's National Action Plan on Climate Change are "walk" and "cycle"

- The MoUD has developed Service Level Benchmarks (SLBs) for urban transport in Indian cities, which integrates the NMT system

Source: (SUTP, 2015)

Non-Motorized Transport in India

Key Policy Initiatives by MoUD in India



- The Jawaharlal Nehru National Urban Renewal Mission (JNNURM), 2005 focused on large scale infrastructure projects including urban transport, required NMT to be made an integral component of the transport system.

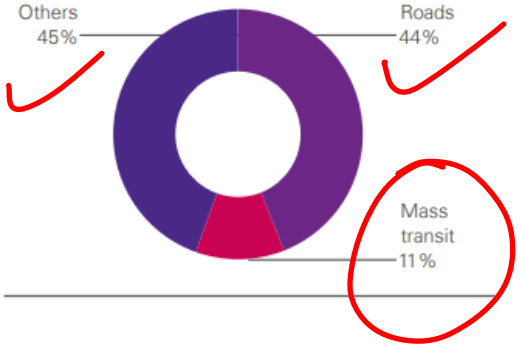
- In 2006, MoUD adopted the NUTP for improving the public transport system and emphasized on developing infrastructure for safe use of NMTs.

- However, NO DIRECT PROJECTS were aimed exclusively for NMTs in India.

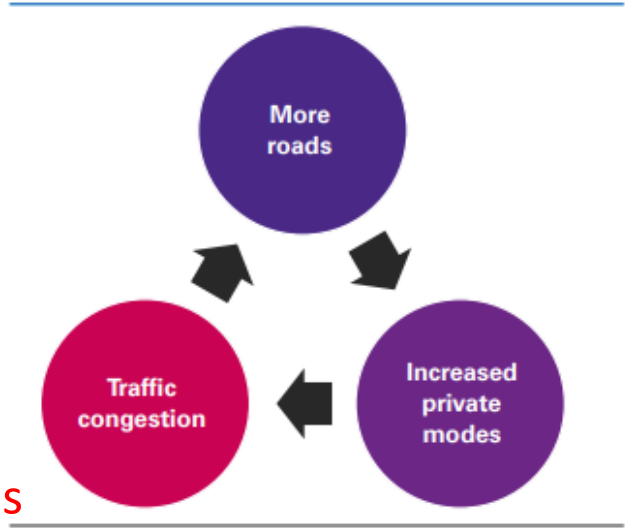
Source: (SUTP, 2015)

Challenges for Public Transport

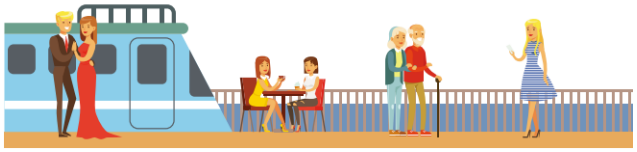
Fund Allocation

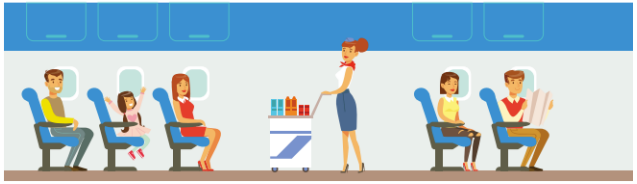


Consequences of investments in roads









Source: (Sustainable smart cities in India: Challenges and future perceptives, Springer, 2017)

Challenges for Public Transport

Benchmarking of Public Transport systems across the Globe

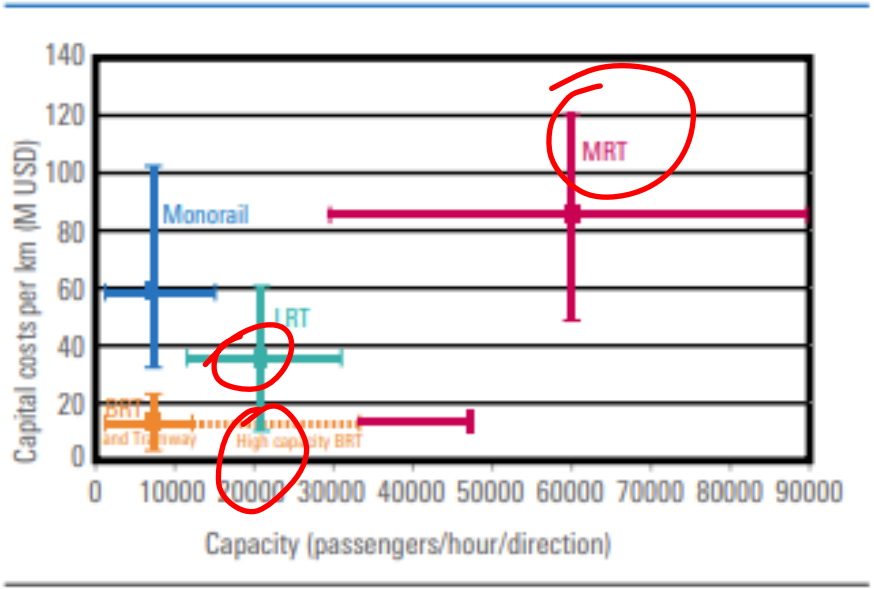
City	Metropolitan area		Bus		MRT		Bus Fleet/1000 persons	MRT Km/1000 persons
	Population ('000)	Area (km²)	Fleet Size	Daily Ridership (mn)	Km	Daily Ridership (mn)		
Beijing	20186	16411	21628	13.8	554	6.7	1.07	0.03
Hong Kong	7184	1104	5743	3.8	178	4	0.80	0.02
Shanghai	23475	6341	16235	7.6	588	6.2	0.69	0.03
Singapore	5312	716	4212	3.5	167	2.2	0.79	0.03
London	8302	1572	7500	6.4	402	3.2	0.90	0.05
New York	8337	790	4344	1.8	370	4.5	0.52	0.04
Seoul	10442	605	7512	4.6	327	6.9	0.72	0.03
Tokyo	13277	2189	1462	0.6	305	2.2	0.11	0.02
Delhi	16788	1483	5942	4.8	215	2.7	0.35	0.01

Indian cities appear to lag behind in the physical coverage of public transport network to most of the cities across the globe

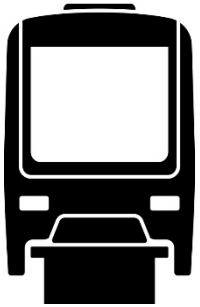
Source: (Sustainable smart cities in India: Challenges and future perceptives, Springer, 2017)

Challenges for Public Transport

Cost and Capacity of Mass Transit Systems



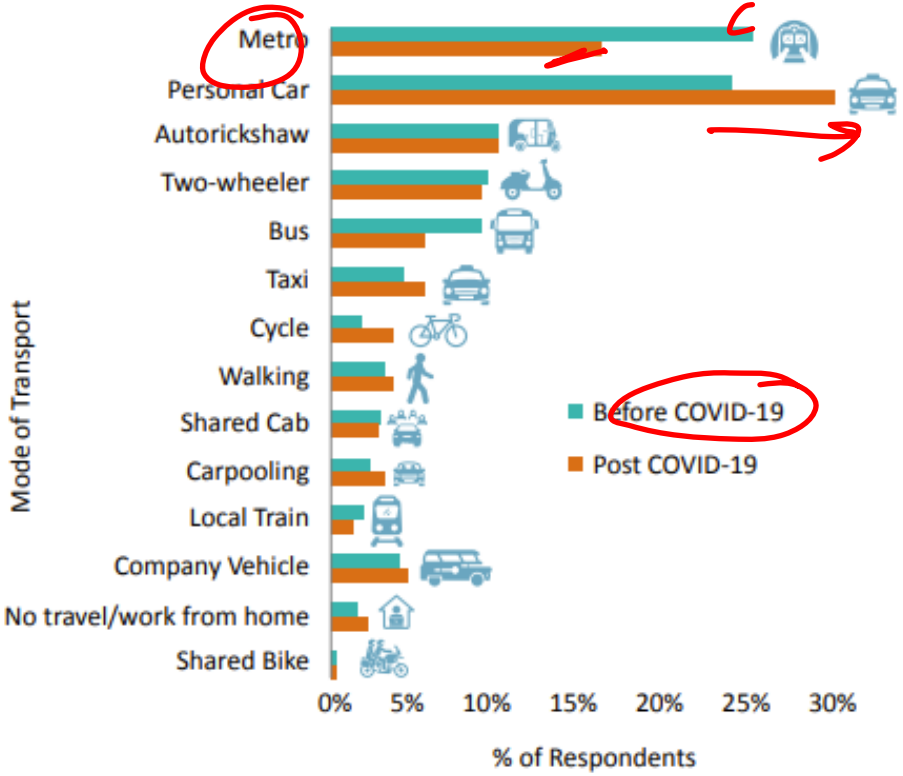






Source: (Sustainable smart cities in India: Challenges and future perceptives, Springer, 2017)

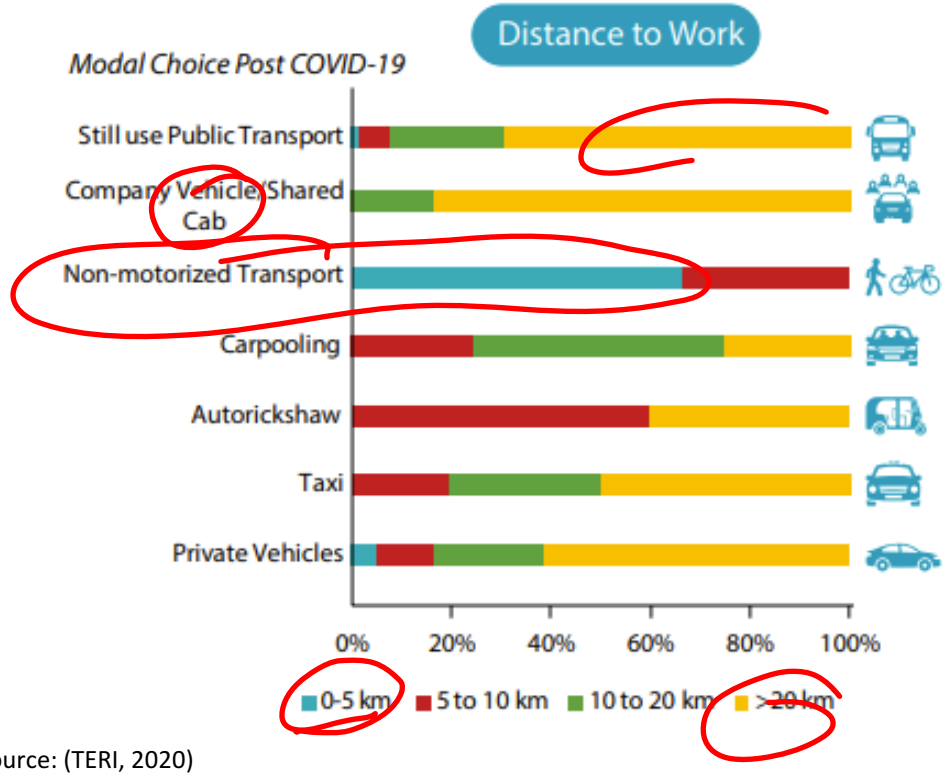
Estimated Modal share Pre and Post COVID-19



- Estimated Modal share Pre and Post COVID-19 for the sample based on TERI survey, April, 2020.

Source: (TERI, 2020)

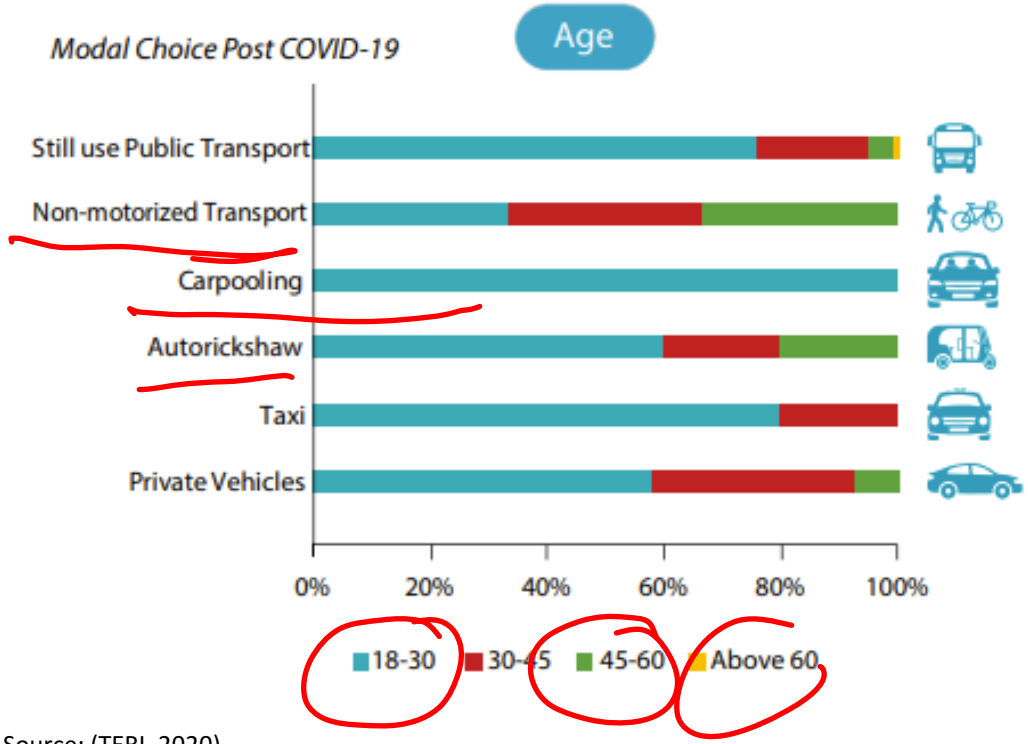
Estimated Modal share Pre and Post COVID-19



Source: (TERI, 2020)

- Distance to work and post-COVID-19 modal choice of Initial public transport users

Estimated Modal share Pre and Post COVID-19



- Age and post-COVID-19 mode choice of Initial public transport users

Source: (TERI, 2020)

Conclusion

- A shift from public transport is expected post-COVID-19.
- Subsequent increase in private vehicles.
- Increase in share of Non-motorized vehicles, especially cycling for short distance trips.
- Scope of NMTs becoming a major short distance travel choice.
- Sustainable measures and policies in India are currently still in the policy levels.



References

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Thank You